Attaching containers to custom networks

In this train, we'll learn how to attach Docker containers to custom networks, enabling precise control over container communication paths.

Learning objectives

- Understand how to attach Docker containers to custom networks, enabling precise control over communication paths.

Attach a container during container creation

If you'd like to attach your container to the custom network when it is created then you can run:

`docker run --network`

Hereby adding the `--network` flag we can specify which network we'd like our container to connect to.

Example:
To start a Python container with the **python:3.9-slim image** and **run app.py**, also attaching the container to the network **python_custom_network**, we can run:
`docker run --network python_custom_network --name python_app python:3.9-slim python app.py`

Attach a running container to a network

In order to attach a running container to a network we can run:

`docker network connect`

Example:
to connect a running container named **python_app** to **python_custom_network**, use:

```
docker network connect python_custom_network python_app
```

Verifying container connections

Once containers are connected to a custom network, you can verify the connections and inspect network settings:

`docker network inspect`

This command will show detailed information about the network, including the connected containers and their IP addresses.

Example:
Using the same network, **python_custom_network** from earlier:
`docker network inspect python_custom_network`

Conclusion

In this train we have learned how to attach containers to custom docker networks.

Key Takeaways

- We can attach containers to custom networks during creation by adding the network flag to `docker run`.
- We can attach running containers to custom networks using `docker network connect`.